

T-Test Challenge Stations

Decide first. Analyze second. Defend your choice.

How this works

Work with your group and move between stations at your own pace.

You are not expected to finish every station.

At each station:

1. Read the entire scenario.
2. Discuss what you think is happening **before running any code**.
3. Decide what analysis or design conclusion you would defend.
4. Use R when the station asks for it.
5. Record your group's reasoning, not just the p-value.
6. If you are unsure, **ask the instructor or TA a specific question**.
7. Move to another open station once your group can explain and defend its answer.

Important: A computer can run a statistical test even when the test makes no scientific sense. Your job today is to decide whether the analysis is defensible.

Station 1

An agency report says that the historical mean dissolved oxygen concentration in a stream was 7.0 mg/L

A biologist samples the stream after restoration and asks:

Has mean dissolved oxygen changed from the historical condition?

At first, one student proposes:

```
t.test(current_do, mu = 7)
```

However, you learn that **7.0 was not a fixed standard**. It was the mean of 12 historical sampling days, and the original historical observations are available.

Use:

station1_dissolved_oxygen.csv

Discuss before coding

1. Is 7.0 a known population value, or is it itself an estimate from a sample?
2. Does that change whether a one-sample t-test is the best analysis?
3. What are the two sources of sampling uncertainty in this study?
4. Which test would your group defend?

Explore the data

```
oxygen <- read.csv("station1_dissolved_oxygen.csv")

aggregate(do_mg_L ~ period,
           data = oxygen,
           FUN = mean)

aggregate(do_mg_L ~ period,
           data = oxygen,
           FUN = sd)
```

Run the test your group thinks is appropriate.

If you choose to compare the two samples directly:

```
t.test(do_mg_L ~ period,
       data = oxygen)
```

Final challenge

Suppose instead that **7.0 mg/L** were a **fixed regulatory threshold**, not an estimated historical mean. Would your choice of t-test change? Explain why.

Station 2

A researcher tests whether a new diet increases fish growth.

There are:

- 4 control tanks
- 4 treatment tanks
- 6 fish in each tank

The diet is assigned to the **tank**, and every fish in a tank receives the same diet.

A student runs:

```
fish <- read.csv("station2_fish_tanks.csv")

t.test(growth_g ~ diet,
       data = fish)
```

and gets a very small p-value.

Your task

Before accepting that result, discuss:

1. What received the treatment: each individual fish or each tank?
2. Are six fish from the same tank six independent treatment replicates?
3. What is the experimental unit?
4. How many independent treatment replicates are actually present?

Now run the student's analysis.

```
fish <- read.csv("station2_fish_tanks.csv")

t.test(growth_g ~ diet,
       data = fish)
```

Then calculate one mean for each tank:

```
tank_means <- aggregate(growth_g ~ diet + tank,
                        data = fish,
                        FUN = mean)

tank_means
```

Now analyze the tank means:

```
t.test(growth_g ~ diet,
       data = tank_means)
```

Discuss

1. Why are the two p-values so different?
 2. Which analysis better matches how the treatment was actually replicated?
 3. If the researcher had used 24 fish per tank instead of 6, would that create more independent treatment replicates?
 4. What design change would actually increase replication?
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Station 3

A researcher measures plant biomass in the **same 12 plots** before and after a management treatment. Unfortunately, the “after” measurement is missing for one plot because a sample was lost.

Use:

station3_paired_missing.csv

Discuss first

1. What makes these data paired?
2. Can Plot 4’s “before” value be paired with Plot 5’s “after” value?
3. What should happen to the plot that has no matching “after” measurement?
4. Would treating all before and after observations as two independent groups preserve the study design?

Look at the data:

```
plants <- read.csv("station3_paired_missing.csv")  
plants
```

Identify complete pairs:

```
complete <- complete.cases(plants$before_biomass_g,  
                             plants$after_biomass_g)  
plants_complete <- plants[complete, ]  
plants_complete
```

Run the paired analysis:

```
t.test(plants_complete$before_biomass_g,  
        plants_complete$after_biomass_g,  
        paired = TRUE)
```

Now, for comparison only, see what happens if someone incorrectly ignores the pairing:

```
t.test(plants$before_biomass_g,  
        plants$after_biomass_g)
```

Final challenge

Why can ignoring the pairing change the result so much?

Station 4

Researchers compare nitrate concentration between a control group and a treatment group.

The groups have different sample sizes, and no previous information tells you that the population variances should be equal.

Use:

station4_variance_choice.csv

Step 1: Examine the data

```
nitrate <- read.csv("station4_variance_choice.csv")

aggregate(nitrate_mg_L ~ group,
          data = nitrate,
          FUN = mean)

aggregate(nitrate_mg_L ~ group,
          data = nitrate,
          FUN = sd)

table(nitrate$group)
```

Discuss:

1. Are the sample sizes equal?
2. Do the sample standard deviations look similar?
3. Were you given a scientific reason to assume equal population variances?

Step 2: Run both versions

Welch:

```
t.test(nitrate_mg_L ~ group,
      data = nitrate)
```

Pooled/equal variance:

```
t.test(nitrate_mg_L ~ group,
      data = nitrate,
      var.equal = TRUE)
```

The important part

The two versions do **not** give the same decision at ($\alpha = 0.05$).

Discuss:

1. Which result should you report?
2. Is it acceptable to choose the version that gives the smaller p-value?
3. Is it acceptable to choose the version that gives the conclusion you expected?
4. What information should determine the test choice?

Information you may use

- The pooled t-test assumes a common population variance.
 - Welch's t-test does not require the two population variances to be equal.
 - You have **not** been given a reason to assume equal variances.
 - In R, Welch's version is the default.
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Station 5

For each scenario, answer two different questions:

1. Could someone technically put these data into `t.test()`?
2. Would a t-test actually answer the scientific question appropriately?

Do not worry about naming the alternative test if you have not learned it yet.

A

Twenty independent irrigated plots and 20 independent non-irrigated plots are compared for **mean corn yield**.

B

Researchers compare whether individual fish **survived or died** between two treatments.

C

The same 18 deer are weighed immediately before and 30 days after a supplemental feeding program.

D

Mean soil pH from 25 independently selected locations is compared with a fixed management target of 6.5.

E

Researchers compare mean plant height among **four fertilizer treatments**.

F

Researchers want to know whether water temperature changes as stream discharge increases.

G

Twenty quadrats are sampled on one north-facing slope and 20 quadrats on one south-facing slope. The researcher wants to conclude that **slope aspect causes differences in plant biomass**.

H

Twenty fish are measured from one restored stream and 20 from one unrestored stream. The researcher wants to conclude that **restoration causes fish to grow larger**.

Station 6

A researcher wants to test whether a new fertilizer increases mean plant height.

The response variable is quantitative, so a two-sample t-test seems reasonable.

The experiment is arranged as follows:

- 5 trays of fertilized plants are placed in Greenhouse A.
- 5 trays of control plants are placed in Greenhouse B.
- Each tray contains 20 plants.
- Greenhouse A is about 4°C warmer.
- Greenhouse A also receives more afternoon light.
- Every plant is measured after six weeks.

The fertilized plants are taller.

Your task

1. What are all the things that changed when fertilizer treatment changed?
2. If plants differ in height, what competing explanations exist?
3. Does having 100 plants in each treatment solve the problem?
4. Does having five trays in each greenhouse solve the problem if treatment is still completely separated by greenhouse?
5. What would you change in the design so that fertilizer is not completely mixed up with greenhouse conditions?

Harder question

Suppose the researcher says:

“But I have 200 observations, so my sample size is huge.”

How would you explain why that statement does not solve the design problem?

Station 7

A field has a strong soil-moisture gradient from one end to the other.

A researcher wants to test whether fertilizer increases crop yield.

Instead of randomly placing six control plots at one end and six fertilized plots at the other, the researcher creates **six nearby pairs of plots** along the gradient.

Within each pair:

- the plots are similar in location and soil,
- one is randomly assigned fertilizer,
- the other is assigned control.

Use:

`station7_matched_plots.csv`

Discuss before coding

1. Why might plots within a pair be more similar than two randomly chosen plots from opposite ends of the field?
2. What source of variation is the pairing trying to control?
3. Are the control and fertilized observations truly unrelated?
4. Which t-test matches this design?

Look at the data:

```
plots <- read.csv("station7_matched_plots.csv")  
  
plots
```

First, pretend you did **not** know about the matching:

```
t.test(plots$control_yield,  
       plots$fertilized_yield)
```

Now use the actual design:

```
t.test(plots$control_yield,  
       plots$fertilized_yield,  
       paired = TRUE)
```

Calculate the pair-by-pair differences:

```
plots$difference <- plots$fertilized_yield -  
                    plots$control_yield  
  
plots  
mean(plots$difference)  
sd(plots$difference)
```

Final challenge

Why is the paired analysis dramatically more informative here?

What would have been lost if the researcher had failed to record which plots formed each pair?

Station 8 — The computer will analyze bad data too

A researcher compares plant height between control and fertilizer treatments.

Use:

station8_check_the_data.csv

A student immediately runs a t-test and reports the result.

Your group should **not** begin with the t-test.

Step 1: Inspect the data

```
plants <- read.csv("station8_check_the_data.csv")

plants

aggregate(height_cm ~ treatment,
            data = plants,
            FUN = mean)

aggregate(height_cm ~ treatment,
            data = plants,
            FUN = sd)
```

Make a quick plot:

```
stripchart(height_cm ~ treatment,
            data = plants,
            vertical = TRUE,
            method = "jitter",
            pch = 19)
```

Discuss

1. Does anything look unusual?
2. What are some possible explanations for an unusual observation?
3. Should you simply delete an observation because it changes the result?
4. What should a researcher do **before** deciding whether a value should be corrected or removed?

If your group finds something that you think needs to be checked, call Dr. Molina or Tara

Do not change the dataset until you have checked with them.

After resolving the data issue

Run the analysis you think is appropriate:

```
t.test(height_cm ~ treatment,  
       data = plants)
```

Final challenge

Explain the difference between:

- deleting a valid extreme observation because you do not like it, and
- correcting a documented data-entry error.

Why is one scientifically unacceptable while the other is necessary?